LABORATORY TEST REPORT

TRUtest DIAGNOSTICS

Name	: **MISS. SAKSHI LONDHE**		
Age/ Gender	: 20 years / Female	Ref. Doctor	: SELF
Sample Type	: SERUM	Collected	: May 12, 2026, 02:19 p.m.
Sample ID	: 260209401	Received	: May 12, 2026, 03:03 p.m.
Client Name	: 2MHPUN129	Reported	: May 12, 2026, 05:51 p.m.

CLINICAL BIOCHEMISTRY

TEST DESCRIPTION	RESULT	UNITS	BIOLOGICAL REFERENCE INTERVAL
Lipid Profile			
Cholesterol - Total (Method: Cholesterol Oxidase, Esterase, peroxidase)	138	mg/dL	<200 : Desirable 200-239 : Borderline risk >240 : High risk
Triglycerides (Method: Lipase / Glycerol Kinase)	61	mg/dL	< 150 :Normal 150–199 :Borderline-High 200–499 :High > 500 :Very High
Cholesterol - HDL (Method: Enzymatic Colorimetric)	58.1	mg/dL	< 40 : Low 40 - 60 : Optimal > 60 : Desirable
Cholesterol - LDL (Method: Enzymatic Colorimetric / Calculated)	67.70	mg/dL	< 100 : Normal 100 - 129 : Desirable 130 – 159 : Borderline-High 160 – 189 : High > 190 : Very High
Cholesterol VLDL (Method: Calculated)	12.20	mg/dL	7 - 40
Cholesterol - Non-HDL (Method: Calculated)	79.90	mg/dL	< 130 Desirable
Total cholesterol / HDL (Method: Calculated)	2.38	Ratio	0 - 5.0
LDL / HDL (Method: Calculated)	1.17	Ratio	0 - 3.5
HDL / LDL (Method: Calculated)	0.86	Ratio	0 - 3.5

Interpretation:

- **For non-fasting samples,** the biological reference interval remains the same for all parameters, except for triglyceride as cholesterol (HDL, LDL, total), which changes only by a small amount in the non-fasting state; **the recommended desired value for triglycerides is <175 mg/dl.** In the non-fasting state, **individuals with a non-fasting triglyceride level >200 mg/dl, are recommended to perform a follow-up fasting lipid panel in 2 to 4 weeks.**
- As per the consensus of the Lipid Association of India, Non-HDL cholesterol and LDL cholesterol can be used as targets to monitor the effectiveness of lipid-lowering therapy.

Associated tests: Apolipoproteins A1, Apolipoproteins B, Apolipoprotein B/A1 Ratio, Lipoprotein(a)

MC-7827

Dr FARHAN SHAIKH
MBBS, MD(Pathology)
R. No: 2013/04/0704

NOTE : This sample is processed at HealthMap Diagnostics Pvt Ltd (Pune) Lab.

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Client Name	: 2MHPUN129	Reported	: May 12, 2026, 05:53 p.m.

CLINICAL BIOCHEMISTRY

TEST DESCRIPTION	RESULT	UNITS	BIOLOGICAL REFERENCE INTERVAL
Liver Function Profile			
Protein Total (Method: Biuret)	7.5	g/dL	6.6 - 8.7
Albumin (Method: Bromcresol Green)	4.8	g/dL	3.5 - 5.2
Globulin (Method: Calculated)	2.70	g/dl	2.5 - 3.5
Albumin / Globulin (Method: Calculated)	1.78	Ratio	1.0 - 2.1
Bilirubin Total (Method: Diazo Method)	**1.4**	mg/dL	0 - 0.9
Bilirubin Direct (Method: Diazo method)	**0.59**	mg/dL	≤ 0.30
Bilirubin Indirect (Method: Calculated)	0.81	mg/dL	up to 1.0
Aspartate Aminotransferase(AST/SGOT) (Method: IFCC Without Pyridoxal Phosphate)	22	U/L	Upto 35
Alanine Transaminase (ALT/SGPT) (Method: UV without pyridoxal -5- phosphate)	13	U/L	Upto 35
Y- Glutamyl Transferase (GGT) (Method: glutamyl-carboxynitroanilide)	10	U/L	<40
Alkaline Phosphatase (ALP) (Method: PNPP, AMP Buffer)	69	U/L	35 - 104
AST / ALT (Method: Calculated)	1.69	Ratio	

Interpretation:

LFT results reflect different aspects of the health of the liver, i.e., hepatocyte integrity(AST & ALT),synthesis and secretion of bile (Bilirubin ,ALP),cholestasis (ALP,GGT),protein synthesis (Albumin).

1. **Hepatocellular injury:**
   - AST-Elevated levels can be seen. However,it is not specific to liver and can be raised in cardiac and skeletal injuries.
   - ALT -Elevated levels indicate hepatocellular damage. It is considered to be most specific lab test for hepatocellular injury.

   Values also correlate well with increasing BMI.
   - Disproportinate increase in AST,ALT compared with ALP. Bilirubin may be elevated. AST: ALT (ratio) - In case of hepatocellular injury AST : ALT >1

   In Alcoholic Liver Disease AST : ALT usually >2

   This ratio is also seen to be increased in NAFLD ,Wilsons's disease, Cirrhosis ,but the increase is usually not >2
2. **Cholestatic pattern:**
   - ALP - Disproportinate increase in ALP compared with AST,ALT. Bilirubin may be elevated. ALP elevation also seen in pregnancy,impacted by age and sex.
   - To establish the hepatic origin correlation with GGT helps. If GGT elevated indicates hepatic cause of increased ALP.

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Sample ID	: 260209401	Received	: May 12, 2026, 03:03 p.m.
Client Name	: 2MHPUN129	Reported	: May 12, 2026, 05:59 p.m.

CLINICAL BIOCHEMISTRY

TEST DESCRIPTION	RESULT	UNITS	BIOLOGICAL REFERENCE INTERVAL
IRON PROFILE			
Iron (Method: Ferrozine – without Deproteinization)	**227**	µg/dL	33 - 193
UIBC (Method: Nitroso-PSAP)	**134**	µg/dL	155 - 355
Iron Binding Capacity - Total (TIBC) (Method: Calculated)	361	µg/dL	240 - 450
Transferrin % (Method: Calculated)	**62.88**	%	20 - 50

Interpretation:

Disease	**Iron**	**TIBC**	**UIBC**	**%Transferrin Saturation**	**Ferritin**
Iron Deficiency	Low	High	High	Low	Low
Hemochromatosis	High	Low	Low	High	High
Chronic Illness	Low	Low	Low/Normal	Low	Normal/High
Hemolytic Anemia	High	Normal/Low	Low/Normal	High	High
Sideroblastic Anemia	Normal/High	Normal/Low	Low/Normal	High	High
Iron Poisoning	High	Normal	Low	High	Normal

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Age/ Gender	: 20 years / Female	Ref. Doctor	: SELF
Sample Type	: WB EDTA	Collected	: May 12, 2026, 02:19 p.m.
Sample ID	: 260209400	Received	: May 12, 2026, 05:25 p.m.
Client Name	: 2MHPUN129	Reported	: May 12, 2026, 06:44 p.m.

HAEMATOLOGY

TEST DESCRIPTION	RESULT	UNITS	BIOLOGICAL REFERENCE INTERVAL
Glycosylated Hemoglobin (GHb/HBA1c)			
HbA1c (Method: HPLC)	5.1	%	Non-Diabetic: <=5.6 Pre Diabetic:5.7-6.4 Diabetic: >=6.5
Estimated Average Glucose (eAG)	99.67	mg/dL	70 - 126

Interpretation

Excellent Control: 6 to 7 %

Fair to Good Control: 7 to 8 %

Unsatisfactory Control: 8 to 10 %

and Poor Control: More than 10%.

Factors Influencing HbA1c Results: Increased levels: Elevated fetal hemoglobin, chronic renal failure, iron deficiency anemia, splenectomy, heightened serum triglycerides, alcohol consumption, poisoning (Lead, Opiate), and salicylate therapy. **Decreased levels:** are often associated with systemic inflammatory diseases and reduced RBC life span, severe iron deficiency & haemolytic anaemia, chronic renal failure, and liver diseases. Clinical correlation reduced red blood cell lifespan (such as in hemolytic anemia or blood loss), following blood transfusions, during pregnancy, excessive intake of Vitamin E or Vitamin C, and hemoglobinopathies, For HbF > 25% and homozygous hemoglobinopathy, an alternate platform (Fructosamine) is recommended for testing of HbA1c.

- HbA1c, also known as glycosylated hemoglobin or glycated hemoglobin, refers to hemoglobin that has glucose molecules attached. It gives an average of glucose levels in the bloodstream for the preceding 2 to 3 months.
- HbA1c has been endorsed by clinical groups & ADA (American Diabetes Association) guidelines for the diagnosis of diabetes using a cut-off point of 6.5%.
- For diabetic patients achieving treatment objectives, the HbA1c test should be conducted at least biannually. If treatment objectives are not met or if a new regimen is initiated, testing once every 3 months is recommended.
- The HbA1c target for non-pregnant adults is generally set at below 7% to prevent microvascular complications.
- In known diabetic patients, the following values can be considered as a tool for monitoring glycemic control.

Associated tests: HOMA IR index, insulin, C-peptide levels.

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Client Name	: 2MHPUN129	Reported	: May 12, 2026, 05:54 p.m.

CLINICAL BIOCHEMISTRY

TEST DESCRIPTION	RESULT	UNITS	BIOLOGICAL REFERENCE INTERVAL
Calcium & Phosphorus			
Calcium (Method: NM-Bapta complex)	9.5	mg/dL	8.6 - 10.3

Interpretation:

The diagnosis and monitoring of a wide range of disorders including diseases of bone, kidney, parathyroid gland, or gastrointestinal tract Calcium levels may also reflect abnormal vitamin D or protein levels. Calcium ions affect the contractility of the heart and the skeletal musculature, and are essential for the function of the nervous system. In addition, calcium ions play an important role in blood clotting and bone mineralization.
Hypocalcemia is due to the absence or impaired function of the parathyroid glands or impaired vitamin-D synthesis. Chronic renal failure is also frequently associated with hypocalcemia due to decreased vitamin-D synthesis as well as hyperphosphatemia and skeletal resistance to the action of parathyroid hormone (PTH). A characteristic symptom of hypocalcemia is latent or manifest tetany and osteomalacia.

TEST DESCRIPTION	RESULT	UNITS	BIOLOGICAL REFERENCE INTERVAL
Phosphorus (Method: Phosphomolybdate - UV)	3.5	mg/dL	2.5 - 4.5

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HAEMATOLOGY

TEST DESCRIPTION	RESULT	UNITS	BIOLOGICAL REFERENCE INTERVAL
Complete Blood Picture (CBP)			
Erythrocytes (EDTA, Whole Blood)			
Hemoglobin (Hb) (Method: Photometry)	13.9	g/dL	12.0 - 15.0
Erythrocyte Count (RBC Count) (Method: Electronic Impedance)	4.58	mil/µL	3.8 - 4.8
Packed Cell Volume(Hematocrit) (Method: Calculated)	42.2	%	36 - 46
Mean Cell Volume (MCV)	92.2	fl	83 – 101
Mean Cell Haemoglobin (MCH)	30.4	pg	27 – 32
Mean Corpuscular Hb Concn (MCHC)	32.9	g/dL	31.5 – 34.5
Red Cell Distribution Width (RDW)-CV	13.1	%	11.6 - 14.5
Red Cell Distribution Width (RDW)-SD	**31.2**	fL	40.0 - 55.0
RBC Morphology			
Remarks	Normocytic Normochromic		
Leucocytes			
Total Leucocyte Count (WBC)	7180	cells/Cumm	4000 - 10000
Neutrophils	50	%	40 - 80
Lymphocytes	**43**	%	20 – 40
Eosinophils	1	%	1 – 6
Monocytes	5	%	2 - 10
Basophils	1	%	0 – 1
Absolute Count			
Absolute Neutrophil Count (Method: Calculated)	3.6	* 10^9/L	2.0 - 7.0
Absolute Lymphocyte Count (Method: Calculated)	3.0	* 10^9/L	1 - 3
Absolute Monocyte Count (Method: Calculated)	0.4	* 10^9/L	0.2 - 1.0
Absolute Eosinophil Count (Method: Calculated)	0.1	* 10^9/L	0.0-0.5
Absolute Basophils Count (Method: Calculated)	0.1	* 10^9/L	0.02-0.10
WBC Morphology	Within Normal Limits		
Platelets			
Platelet Count (Method: Electronic Impedance)	274	10^3/uL	150 - 450
Mean Platelet Volume (MPV)	8.5	fL	7.2 - 11.7
Platelet Morphology	Adequate on smear		
PDW	13.7	%	9 - 17

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HAEMATOLOGY

TEST DESCRIPTION	RESULT	UNITS	BIOLOGICAL REFERENCE INTERVAL
PCT (Method: Calculated)	0.23	%	0.2 - 0.5

Tests done on Automated Five Part Cell Counter. (WBC, RBC,Platelet count by impedance method, colorimetric method for Hemoglobin, WBC differential by flow cytometry using laser technology other parameters are calculated). All Abnormal Haemograms are reviewed confirmed microscopically.

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Sample ID	: 260209401	Received	: May 12, 2026, 03:03 p.m.
Client Name	: 2MHPUN129	Reported	: May 12, 2026, 05:26 p.m.

CLINICAL BIOCHEMISTRY

TEST DESCRIPTION	RESULT	UNITS	BIOLOGICAL REFERENCE INTERVAL
Thyroid Profile-II			
Triiodothyronine Total (TT3) (Method: ECLIA)	143.00	ng/dL	91 - 218
Triiodothyronine Free (FT3) (Method: ECLIA)	2.65	pg/mL	2.56 - 5.01
Thyroxine - Total (TT4) (Method: ECLIA)	10.9	ug/dL	5.91 - 13.2
Thyroxine - Free (FT4) (Method: ECLIA)	1.33	ng/dL	0.98 - 1.63
Thyroid Stimulating Hormone (TSH) (Method: ECLIA)	1.24	uIU/mL	0.51- 4.30

Interpretation:

Important Note:

A TSH value of up to 15 µIU/ml needs clinical correlation or repeat testing with a new sample, as physiological factors can give falsely high TSH. Transiently Raised TSH can occur due to non-thyroid illnesses like severe infections liver, cardiac, and kidney diseases, burns, surgery, and trauma.

Diurnal Variability: TSH Follows a Diurinal rhythm and is at maximum between 2 am to 4 am and a minimum between 6 to 10 pm. Variation is about 50-206%.

Limitations:

- Patients receiving High Biotin Dose treatment should have a gap of a minimum of 8 hrs, before giving the blood sample.
- Heterophile Antibodies can react in Immunoassay procedures giving erroneous results. The assay is designed to minimize heterophile antibody interference.

Please Note: FT3 and FT4 Measure the free "unbound" fraction of the T3 and T4 and are considered more sensitive compared to the Total T3 and T4 to assess thyroid status.

Associated tests: Anti-thyroid Antibodies, USG thyroid, TSH receptor Antibody. Thyroglobulin, Calcitonin.

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CLINICAL BIOCHEMISTRY

TEST DESCRIPTION	RESULT	UNITS	BIOLOGICAL REFERENCE INTERVAL
Kidney Basic screen			
Creatinine (Method: Jaffe Kinetic)	0.6	mg/dL	0.50 - 0.90
Urea (Method: Urease)	17.2	mg/dL	16.6 - 48.5
Blood Urea Nitrogen (BUN) (Method: Calculated)	8.04	mg/dL	8 - 23
Blood Urea Nitrogen (BUN)/Creatinine (Method: Calculated)	13.40	Ratio	6 - 22
Sodium (Method: ISE)	139.6	mmol/L	136 - 145
Potassium (Method: ISE)	4.26	mmol/L	3.5 - 5.1
Chloride (Method: ISE)	100.2	mmol/L	98 - 107
Uric Acid (Method: mg/dL)	3.5	mg/dL	2.4 - 5.7
Glomerular Filtration Rate (eGFR) (Method: Calculated)	**131.70**	mL/min	90 - 120 mL/min/1.73 m2
Urea/Creatinine Ratio (Method: Calculated)	28.67		20 - 35

Interpretation:

Creatinine: Muscles produce creatinine, a waste product, from creatine phosphate, a substance that stores a lot of energy. Unlike urea, the amount of creatinine generated is constant and mostly depends on muscle mass. Age, gender, race, muscularity, exercise, pregnancy, and several other physiulogical characteristics can all have an impact on serum creatinine levels. Decreased serum Creatinine is associated with increasing Age and poor muscle mass, such as muscular atrophy. Both acute and chronic renal disease and blockage are associated with elevated blood creatinine levels. Creatinine is not an appropriate indicator for identifying kidney disease in its early stages since an increase in blood creatinine is only seen when there is significant nephron damage.

High **Urea, Uric Acid, and Blood Urea Nitrogen (BUN)** could indicate poor renal function, in addition to other etiulogies

Sodium:

- **Low levels:** prulonged vomiting or diarrhea, diminished reabsorption in the kidney, and excessive fluid retention. **Pseudo-hyponatremia** is **a laboratory artifact**. It is usually caused by hypertriglyceridemia, chulestasis (lipoprotein X), and hyperproteinemia (monoclonal gammopathy, intravenous immunoglobulin [IVIG]). **Diluted sampling** should also be suspected in such cases and confirmed on a repeat fresh sample if indicated clinically.
- **High levels:** excessive fluid loss, high salt intake, and increased kidney reabsorption.

Potassium:

- **Low levels:** reduced intake of dietary potassium or excessive loss of potassium from the body due to diarrhea, prulonged vomiting, or increased renal excretion.
- **High levels:** dehydration or shock, severe burns, hemulysis, diabetic ketoacidosis, and retention of potassium by the kidney. **Pseudohyperkalemia, which may result from a hemulyzed or aged sample, should always be ruled out by doing a repeat electrulyte estimation on a fresh sample, as clinically indicated.**

Chloride:

- **Low levels** are noted in reduced dietary intake, prulonged vomiting, and reduced renal reabsorption, as well as some forms of acidosis and alkalosis.
- **High levels** are found in dehydration, kidney failure, some forms of acidosis, high dietary or parenteral chloride intake, and salicylate poisoning

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